NMBS-C7447 at $z = 1.80$ in CPTG

Native-Phase Placement, Cross-Scale Locking, and Structural Mode N

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Abstract

NMBS-C7447 is a massive compact quiescent galaxy at redshift $z = 1.80$ whose combination of high stellar mass, small physical size, and large stellar velocity dispersion made it an important early dynamical demonstration of the strong structural evolution of quiescent galaxies. The source study reports a stellar mass of order $1.5 \times 10^{11} M_{\odot}$, circularized effective radius $R_e = 1.64 \pm 0.15\,\mathrm{kpc}$, and stellar velocity dispersion $\sigma_{\star} = 294 \pm 51\,\mathrm{km\,s^{-1}}$. This paper uses those measurements to test whether the galaxy's independently inferred gravitational structure is related to the native geometric scale of Curvature Polarization Transport Gravity (CPTG) at the same cosmological phase. In the declared one-radius pressure-supported reduction, the observations give $a_{\star} \simeq a_{\sigma} = (1.708 \pm 0.613) \times 10^{-9}\,\mathrm{m\,s^{-2}}$, while the CPTG native π branch gives $a_{\pi}(z = 1.80) = 1.64863 \times 10^{-9}\,\mathrm{m\,s^{-2}}$. The central values differ by 3.60%, corresponding to only $0.097 \simeq 0.10\sigma$ relative to the propagated observational uncertainty and to the cross-scale ratio $\eta_{\star} = a_{\star}/a_{\pi} = 1.036 \pm 0.372$. The near-unit result persists across the current CPTG acoustic-to-luminosity distance-coordinate sensitivity. The analysis then uses the measured Sérsic structure and flattening to reconstruct the radial CPTG field, yielding a measured-flattening Structural Mode $N = 3.06465$ with explicit source and geometry sensitivity bounds. The two quantities address different questions: $\eta_{\star}$ measures cross-scale placement relative to native geometry, whereas N measures the internal radial organization of the solved field. The result establishes a well-defined single-object consistency case and motivates a predeclared multi-object test of native locking across independent native phases.

1 NMBS-C7447 as a Cosmological and Dynamical Test Case

NMBS-C7447 is a compact, massive quiescent galaxy at $z = 1.80$ whose combination of stellar mass, size, and velocity dispersion makes it an unusually informative link between early galaxy structure and cosmological interpretation. Its stars are already confined to a remarkably small volume, while its spectrum shows little or no ongoing star formation. The original VLT X-Shooter and HST-WFC3 analysis found a stellar mass of order $1.5 \times 10^{11} M_{\odot}$, a young best-fit stellar population age of about $0.40\,\mathrm{Gyr}$, a circularized effective radius of only $1.64 \pm 0.15\,\mathrm{kpc}$, and an aperture-corrected stellar velocity dispersion of $294 \pm 51\,\mathrm{km\,s^{-1}}$ [1]. By the observed epoch, a very large stellar mass had therefore already assembled into a dynamically hot, compact system and largely ceased forming new stars.

That combination made NMBS-C7447 important well before CPTG was applied to it. The source study showed that, at fixed dynamical mass, the galaxy is about 2.5 times smaller than comparable low-redshift systems and has a stellar velocity dispersion about 1.8 times higher [1]. Those measurements helped establish that the compactness of high-redshift quiescent galaxies was not merely a photometric artifact: their stellar dynamics independently confirmed that these systems were genuinely denser and structurally different from their present-day counterparts.

For cosmology and galaxy formation, the importance of NMBS-C7447 lies in the combination of early epoch and mature internal structure. A successful framework must accommodate both the existence of such a compact quenched system and the relation between its bound gravitational state and the cosmological geometry in which it is observed. In the standard ΛCDM picture, the redshift is placed on a Friedmann expansion history and galaxy assembly is interpreted within hierarchical structure growth in dark-matter halos; the source study places the compactness and subsequent size evolution of systems such as NMBS-C7447 in that broader hierarchical context [2, 1]. For this object, however, van de Sande et al. do not obtain the quoted dynamical mass from a fitted dark-halo profile: it is a Sérsic-dependent virial estimate constructed from the measured dispersion and radius. CPTG uses the same measurements to test a different relation between bound-system structure and cosmological geometry.

1.1 Observational Leverage for a CPTG Test

For the present analysis, NMBS-C7447 is valuable because the observations constrain cosmological placement and bound-system structure through distinct measurements. Its redshift tells us when the light was emitted. Its stellar velocity dispersion and effective radius tell us how strongly the stellar body is gravitationally supported. Its Sérsic profile and projected axis ratio tell us how the stellar source is distributed in radius and shape. Together these observables permit the cosmological and structural parts of the CPTG calculation to be constructed separately before they are compared.

The source measurements used here are

$$\sigma_\star = 294 \pm 51\,\mathrm{km\,s^{-1}}, \qquad R_e = 1.64 \pm 0.15\,\mathrm{kpc}, \tag{1}$$

with Sérsic index $n = 5.3 \pm 0.4$, projected axis ratio $b/a = 0.71 \pm 0.01$, and stellar mass of order $1.5 \times 10^{11} M_\odot$ [1]. The published dynamical mass, $(1.7 \pm 0.5) \times 10^{11} M_\odot$, is useful as a source-level consistency check, but it is derived from the same dispersion, radius, and Sérsic-dependent virial coefficient and is therefore not an independent CPTG validation target.

2 CPTG Quantities and Test Architecture

The analysis relies on three CPTG quantities that describe different aspects of the same observed system: the object-dependent structural acceleration $a_\star$, the native cosmological acceleration a_π, and the post-solution Structural Mode N. Their roles must remain distinct because each is obtained from a different stage of the inference.

2.1 Object-Dependent Structural Acceleration $a_\star$

In CPTG, $a_\star$ is the structural acceleration scale associated with a gravitating object. It is not a universal acceleration constant shared by all galaxies, and it is not simply the acceleration at one arbitrary radius. It is the object-dependent scale that enters the curvature-polarization/transport response of the solved system [3]. Different observing configurations can estimate the same physical quantity in different ways. For a pressure-dominated galaxy with only an integrated velocity dispersion and an effective radius, the declared one-radius estimator used here is

$$a_\sigma = \frac{\sigma_\star^2}{R_e}. \tag{2}$$

This tells us, at the simplest effective-radius level, the acceleration scale implied by the observed stellar support. The structural estimate is therefore supplied by the galaxy data themselves; it is not assigned by the cosmological branch.

The same structural scale has a CPTG curvature-radius representation,

$$R_c = \frac{48\pi c^2}{a_\star}, \tag{3}$$

where lowercase R_c denotes a bound-system structural curvature scale, not the physical radius of the visible galaxy.

2.2 Native Cosmological Acceleration a_π

CPTG also has one native geometric π branch that describes cosmological phase. At any redshift, that branch supplies a universe-scale acceleration coordinate $a_\pi(a)$ and corresponding uppercase curvature-transport radius

$$R_C(a) = \frac{48\pi c^2}{a_\pi(a)}. \tag{4}$$

Thus $a_\star$ characterizes the bound galaxy, whereas a_π characterizes the native cosmological geometry. The two scales are derived separately and compared only after both have been fixed.

This separation is one of the ways CPTG differs from the ΛCDM construction used to report the original physical galaxy dimensions. In this paper, a flat-ΛCDM expansion function is not used to generate or rescale $a_\star$. Redshift first selects the native CPTG phase. The observed galaxy then supplies its own structural acceleration. The test asks whether those independently obtained scales have a reproducible geometric relation.

2.3 Cross-Scale Native-Locking Coordinate

Because R_c and R_C contain the same geometric numerator, CPTG admits the dimensionless cross-scale coordinate

$$\eta_\star \equiv \frac{R_C}{R_c} = \frac{a_\star}{a_\pi}. \tag{5}$$

The candidate native-locking hypothesis asks whether this dimensionless ratio is preserved when the same structural state is represented along native phase. The strongest and simplest benchmark is unit locking,

$$\eta_\star = 1, \tag{6}$$

which means that the measured bound-system structural scale and the native cosmological scale coincide at that phase. This value is not fitted to NMBS-C7447. In the declared one-radius reduction, the galaxy supplies an estimate of $a_\star$ from its measured support; the CPTG branch determines a_π from z; only then are the two numbers compared.

2.4 Post-Solution Structural Mode N

The cross-scale ratio characterizes the placement of the galaxy relative to native geometry, but it does not describe the internal organization of the solved field. Structural Mode N provides that second, downstream description. Structural Mode is evaluated only after the baryonic source, $a_\star$, and nonlinear CPTG field have been fixed. It measures where the radial curvature-polarization/transport structure changes most strongly. Lower N corresponds to a response whose dominant structural variation is weighted farther outward; higher N corresponds to a response organized progressively farther inward [5]. It is therefore a structural diagnostic, not another free fitting parameter.

The distinction between the two N symbols used in this paper will matter throughout:

$$N_\pi: \text{ cosmological phase,} \qquad N: \text{ post-solution radial Structural Mode.} \tag{7}$$

The first tells us *when* the galaxy is represented on the native branch. The second tells us *where* the solved field is organized inside the galaxy.

2.5 Separation of Observables and Derived Coordinates

The test is constructed to preserve the separation between observational inputs and derived coordinates. Redshift fixes the native cosmological phase without determining the galaxy's structural acceleration; the galaxy measurements supply the declared one-radius estimate of $a_\star$ without setting the native phase; and Structural Mode is evaluated only after the radial field has been solved. The comparison is therefore made after the relevant quantities have been obtained rather than being imposed as part of their definitions.

With that architecture established, the central numerical comparison can be previewed before its derivation:

$$a_\sigma^{\mathrm{obs}} = (1.70805 \pm 0.61284) \times 10^{-9}\,\mathrm{m\,s^{-2}}, \tag{8}$$

$$a_\pi^{\mathrm{CPTG}}(z = 1.80) = 1.64863 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{9}$$

Their central difference is

$$a_\sigma^{\mathrm{obs}} - a_\pi^{\mathrm{CPTG}} = 5.94 \times 10^{-11}\,\mathrm{m\,s^{-2}}, \tag{10}$$

or

$$\frac{a_\sigma^{\mathrm{obs}} - a_\pi^{\mathrm{CPTG}}}{a_\pi^{\mathrm{CPTG}}} = 3.60\%. \tag{11}$$

Measured against the observational uncertainty in the pressure-support scale, the separation is

$$\frac{a_\sigma^{\mathrm{obs}} - a_\pi^{\mathrm{CPTG}}}{\delta a_\sigma} = 0.09696 \simeq 0.10\sigma. \tag{12}$$

The central values are close, while the present observational uncertainty remains substantially larger than their separation. The correct interpretation is not that one galaxy has proved native locking. It is that the simplest unit-lock benchmark lands well inside the observed support interval without having been used to determine that interval. The rest of the paper shows exactly how each side of this comparison is constructed and then asks whether the solved galaxy field contains additional structural information through N.

3 Observational Inputs and Reduction Strategy

Table 1 collects the observational quantities that enter the analysis and distinguishes them from the quantities derived within CPTG. Maintaining that distinction is central to the interpretation of the results. The primary calculation retains the published physical quantities under the source paper's original distance convention; a later section asks how much the answer moves if those distance-dependent quantities are re-expressed through the current CPTG comparison coordinates. That later exercise is a sensitivity test, not a hidden recalibration of the primary result.

Quantity	Adopted value	Role in this paper
Spectroscopic redshift	$z = 1.80$	Selects the common CPTG native phase
Aperture-corrected stellar dispersion	$294 \pm 51\,\mathrm{km\,s^{-1}}$	Pressure-support measurement
Circularized F160W effective radius	$1.64 \pm 0.15\,\mathrm{kpc}$	One-radius support scale; area-equivalent Sérsic baseline
Sérsic index	$n = 5.3 \pm 0.4$	Radial stellar-source shape
Projected axis ratio	$q_{\rm proj} = b/a = 0.71 \pm 0.01$	Oblate source-geometry constraint
Stellar mass	$M_\star \simeq 1.5 \times 10^{11} M_\odot$	Radial source normalization; source paper adopts ~ 0.2 dex systematic uncertainty
Published dynamical mass	$(1.7 \pm 0.5) \times 10^{11} M_\odot$	Shared-input virial consistency reference, not an independent validation observable

Table 1: Observed NMBS-C7447 quantities used in the native-locking and Structural Mode calculations [1].

Table 2 summarizes the four-stage reduction. Each stage addresses a distinct physical question, and later derived quantities are not used to redefine earlier inputs.

Stage	Observational information	CPTG output	Physical question
1	Redshift z	a, N_π, E_π, a_π, R_C	Where is the observation on the native cosmological geometry?
2	$\sigma_\star$ and R_e	$a_\sigma \to a_\star$ and R_c	What structural acceleration is required by the observed stellar support?
3	Outputs of Stages 1 and 2	$\eta_\star = a_\star/a_\pi = R_C/R_c$	Does the bound-system scale align with the native scale?
4	$M_\star$, n, $q_{\rm proj}$, and solved $g(r)$	λ and Structural Mode N	Where is the returned CPTG field organized inside the galaxy?

Table 2: Inference sequence for the NMBS-C7447 test. The ordering keeps native-phase placement, bound-system structural inference, cross-scale comparison, and post-solution Structural Mode logically distinct.

The sequence begins with cosmological placement from redshift and then turns to the independently observed stellar system. The declared one-radius structural estimate is fixed before $\eta_\star$ is formed, and the radial field is solved before N is evaluated. This ordering is what makes the central 3.60% comparison informative: the two acceleration scales meet at the end of the calculation rather than being made equal at the beginning.

4 Native-Phase Placement of NMBS-C7447

The cosmological part of the calculation begins with the measured redshift alone. Galaxy mass, radius, dispersion, Sérsic structure, and Structural Mode do not enter this stage. CPTG uses one native geometric π branch, so $z = 1.80$ selects a unique native phase before the galaxy itself is solved. The complementary geometric weights are [3, 4]

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \qquad \mathcal{T}_\pi = \frac{1}{\pi}, \qquad \mathcal{B}_\pi + \mathcal{T}_\pi = 1, \tag{13}$$

with native expansion function

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}. \tag{14}$$

The observed redshift selects

$$a = \frac{1}{1+z} = 0.3571428571, \qquad N_\pi \equiv \ln a = -1.029619417. \tag{15}$$

At this phase,

$$E_\pi(1.80) = 2.960729196. \tag{16}$$

The same native phase determines the corresponding cosmological acceleration scale. The parent CPTG framework fixes its present anchor geometrically; it is not obtained from NMBS-C7447 or from a fit to the galaxy. The native present acceleration anchor is

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} = 5.568327997 \times 10^{-10}\,\mathrm{m\,s^{-2}}, \tag{17}$$

so the native acceleration at the observed phase is

$$a_\pi(z) = a_{\pi,0} E_\pi(z) = 1.648631127 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{18}$$

This is the cosmological-side number that will eventually be compared with the galaxy's independently measured structural acceleration. In radius language, the same native state is represented by

$$R_C(z) = \frac{48\pi c^2}{a_\pi(z)} = 266.4149\,\mathrm{Gpc}. \tag{19}$$

The native rate is

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a), \qquad H_0^{(\pi)} = 69.4162507897\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{20}$$

with the equivalent acceleration form

$$H_\pi(a) = \frac{a_\pi(a)}{c\sqrt{\mathcal{B}_\pi}}. \tag{21}$$

For a branch of the form in Eq. (14), direct integration of $dt = da/[aH_\pi(a)]$ gives

$$t_\pi(a) = \frac{2}{\pi H_0^{(\pi)} \sqrt{\mathcal{B}_\pi}} \sinh^{-1}\left[\sqrt{\frac{\mathcal{B}_\pi}{\mathcal{T}_\pi}}\, a^{\pi/2}\right]. \tag{22}$$

At the measured phase this gives

$$t_\pi(z = 1.80) = 3.11117\,\mathrm{Gyr}, \qquad t_{\pi,0} = 12.75412\,\mathrm{Gyr}. \tag{23}$$

These clock values locate the observation on the native history; they do not determine the galaxy's local $a_\star$. This distinction is deliberate. In a standard ΛCDM reduction, the same redshift would normally be converted to time and distance using a Friedmann background. Here the governing CPTG phase comes from the native π branch, while the original source cosmology is retained only where needed to respect how the published physical radius and mass were reported. The source redshift is treated at its published value $z = 1.80$ because no redshift uncertainty is used in the source reduction; the observational uncertainty budget below is therefore dominated by $\sigma_\star$ and R_e.

5 Pressure-Supported Structural Acceleration

Once the native phase has been fixed independently, the analysis turns to the bound galaxy. NMBS-C7447 is pressure supported in the present reduction, so the measured velocity dispersion tells us how rapidly stars move in the gravitational potential and the effective radius supplies the characteristic spatial scale over which that support is measured. For the present one-radius reduction, the relevant support coordinate is

$$a_\sigma = \frac{\sigma_\star^2}{R_e}. \tag{24}$$

Dimensionally, $\sigma_\star^2/R_e$ is an acceleration. Physically, it is a one-radius summary of the gravitational support demanded by the observed stellar motions. CPTG does not identify a_σ with $a_\star$ by definition in every system. In this explicitly declared pressure-dominated one-radius reduction, a_σ is adopted as the estimator of the object's structural acceleration scale $a_\star$. The available integrated kinematics do not separately resolve ordered rotation, inclination, or orbital anisotropy, so those support-decomposition uncertainties remain outside the scalar estimator. A future resolved Jeans analysis could replace this one-radius estimate with a fuller radial determination without changing what $a_\star$ means.

Using the published midpoint values gives

$$a_\star \simeq a_\sigma = \frac{(294\,\mathrm{km\,s^{-1}})^2}{1.64\,\mathrm{kpc}} = 1.708048772 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{25}$$

For independent quoted errors in $\sigma_\star$ and R_e,

$$\left(\frac{\delta a_\sigma}{a_\sigma}\right)^2 = \left(2\frac{\delta\sigma_\star}{\sigma_\star}\right)^2 + \left(\frac{\delta R_e}{R_e}\right)^2, \tag{26}$$

which gives

$$a_\star \simeq a_\sigma = (1.708 \pm 0.613) \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{27}$$

This provides the galaxy-side structural quantity for the later cross-scale comparison. The uncertainty is large—about 36%—because a_σ depends quadratically on the measured dispersion, and the present $\pm 51\,\mathrm{km\,s^{-1}}$ dispersion error dominates the budget. That fact will matter when the apparent closeness to unit locking is interpreted.

The same local structural state can be written as a CPTG curvature radius,

$$R_c = \frac{48\pi c^2}{a_\star} = 257.1471\,\mathrm{Gpc}. \tag{28}$$

Under the same linearized independent-error propagation used for a_σ, its observational scale is

$$R_c = 257.15 \pm 92.26\,\mathrm{Gpc}. \tag{29}$$

This very large radius should not be pictured as the visible size of the galaxy. It is the curvature-transport scale associated with the object's structural acceleration in the CPTG parameterization. The visible stellar body is still measured in kiloparsecs; R_c is a derived geometric scale.

6 Cross-Scale Native-Locking Construction

The cosmological and bound-system calculations now meet. The native branch supplies a_π from redshift, while the galaxy supplies the declared one-radius estimate of $a_\star$ from stellar support, with neither quantity used to determine the other. CPTG can compare them because their associated curvature radii belong to the same geometric family. The natural dimensionless ratio is

$$\eta_\star(a) \equiv \frac{R_C(a)}{R_c(a)}. \tag{30}$$

Using Eqs. (3) and (4), the common geometric numerator cancels exactly:

$$\eta_\star(a) = \frac{a_\star(a)}{a_\pi(a)}. \tag{31}$$

This cancellation is exact. No observational Hubble parameter, matter fraction, dark-energy fraction, or galaxy halo parameter is required to obtain the identity itself. What remains a hypothesis is not the algebra but the proposed behavior of the ratio across native phase.

A native-locked structural trajectory is the special case

$$\eta_\star(a) = \eta_\star = \text{constant}. \tag{32}$$

Because

$$a_\pi(a) = a_{\pi,0} E_\pi(a), \tag{33}$$

Eq. (32) implies

$$a_\star(a) = \eta_\star a_{\pi,0} E_\pi(a). \tag{34}$$

Equivalently, the phase-normalized structural coordinate

$$\widehat{a}_\star \equiv \frac{a_\star(a)}{E_\pi(a)} \tag{35}$$

is invariant along the comparison trajectory:

$$\widehat{a}_\star = \eta_\star a_{\pi,0}. \tag{36}$$

The locking hypothesis is intentionally narrower than a universal assertion that $a_\star = a_\pi$ for every object. Different bound systems can have different $a_\star$ because their structure is different. The constant ratio $\eta_\star$ carries that object dependence. Nor does the construction claim that NMBS-C7447 will literally preserve its present internal structure as the universe evolves. It defines a comparison trajectory: if the same structural state is transported along native phase, does its ratio to the native geometric scale remain fixed? Unit locking, $\eta_\star = 1$, is simply the strongest benchmark inside that broader hypothesis.

The same hypothesis can be written exactly in rate form using

$$H_{\mathrm{struct}} = \frac{16\pi}{3}\frac{a_\star}{c}. \tag{37}$$

Combining this with Eq. (21) gives

$$\mathcal{K}_\star \equiv \frac{H_{\mathrm{struct}}}{H_\pi} = \frac{16\pi}{3}\sqrt{\mathcal{B}_\pi}\,\eta_\star. \tag{38}$$

For NMBS-C7447,

$$\mathcal{K}_\star = 14.332 \pm 5.142, \tag{39}$$

where the uncertainty is inherited from the same $\sigma_\star$ and R_e propagation as $\eta_\star$. This rate form is mathematically equivalent to the acceleration/radius form; it is included to expose the connection, not to introduce a second independent test. The radius/acceleration formulation remains primary because the common $48\pi c^2$ numerator makes the geometric content transparent.

7 Native-Locking Result for NMBS-C7447

Applying the cross-scale ratio to NMBS-C7447 gives the principal result of the paper. The galaxy-side acceleration is $(1.708 \pm 0.613) \times 10^{-9}\,\mathrm{m\,s^{-2}}$; the native CPTG value at the measured redshift is $1.64863 \times 10^{-9}\,\mathrm{m\,s^{-2}}$. Their midpoint difference is only 3.60%, far smaller than the present observational uncertainty. In the geometry's natural dimensionless form, Eqs. (18) and (25) give

$$\eta_\star = \frac{a_\star}{a_\pi} = \frac{1.708048772 \times 10^{-9}}{1.648631127 \times 10^{-9}} = 1.036040594. \tag{40}$$

Equivalently, the radius representation gives the same algebraic ratio,

$$\frac{R_C}{R_c} = \frac{266.4149}{257.1471} = 1.036040594. \tag{41}$$

Propagating the quoted dispersion and radius uncertainties gives

$$\eta_\star = 1.036 \pm 0.372. \tag{42}$$

The result is therefore centered slightly above exact unit locking but is observationally indistinguishable from it at the present precision. Quantitatively, the unit-lock benchmark $\eta_\star = 1$ differs from the measured midpoint by only

$$\frac{\eta_\star - 1}{\delta\eta_\star} = 0.097, \tag{43}$$

or approximately 0.10σ under the stated independent-error approximation. The residual is therefore small compared with the observational error budget: the unit-lock benchmark differs from the measured central support scale by much less than one propagated standard deviation. This is a consistency result, not a likelihood ratio or proof of the hypothesis. The small residual records the closeness of the central values; the approximately 36% observational uncertainty means that the present measurement does not yet discriminate tightly among ratios around unity. The direct signed-corner bracket is

$$0.648 \leq \eta_\star \leq 1.570, \tag{44}$$

which is a sensitivity interval rather than a confidence interval.

The phase-normalized structural coordinate is

$$\widehat{a}_\star = \frac{a_\star}{E_\pi} = (5.769 \pm 2.070) \times 10^{-10}\,\mathrm{m\,s^{-2}}, \tag{45}$$

compared with

$$a_{\pi,0} = 5.56833 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{46}$$

The corresponding native-locked present-coordinate structural radius would be

$$R_{c,0}^{\mathrm{lock}} = R_c(z)E_\pi(z) = 761.3431\,\mathrm{Gpc}, \tag{47}$$

with linearized observational scale

$$R_{c,0}^{\mathrm{lock}} = 761.34 \pm 273.16\,\mathrm{Gpc}. \tag{48}$$

This is a comparison coordinate implied by the lock hypothesis, not a prediction of the literal future radius of the evolving galaxy. It represents the same locked structural state at the normalized present phase and should not be interpreted as a forecast of the galaxy's actual future evolution.

7.1 Dependence on the Source Distance Convention

The published radius and stellar mass are not raw angular quantities; they were converted to physical units using the source paper's flat-ΛCDM convention, $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$ [1]. Because these physical quantities inherit that distance convention, the robustness of the native-locking result should also be checked against the current CPTG comparison coordinates. The primary result deliberately preserves the published measurements. The following calculation changes only the distance coordinate as a stress test.

For any comparison branch b, the angular-diameter coordinate used here is

$$D_{A,b}(z) = \frac{c}{(1+z)H_{0,b}} \int_0^z \frac{dz'}{E_b(z')}. \tag{49}$$

The current acoustic comparison coordinate uses

$$p_{\mathrm{ac}} = 3 - \frac{\pi}{100}, \qquad E_{\mathrm{ac}}^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\mathrm{ac}}}, \qquad H_{0,\mathrm{ac}} = 67.4777967351\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{50}$$

while the luminosity comparison coordinate uses

$$\Omega_{m,\mathrm{ac}} = \frac{p_{\mathrm{ac}}}{3\pi}, \qquad \Omega_{\Lambda,\mathrm{ac}} = 1 - \Omega_{m,\mathrm{ac}}, \qquad E^2_{\mathrm{lum}} = \Omega_{\Lambda,\mathrm{ac}} + \Omega_{m,\mathrm{ac}} a^{-p_{\mathrm{ac}}}, \tag{51}$$

with $H_{0,\mathrm{lum}} = 73.17\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. These are comparison coordinates evaluated at the same native phase, not additional native cosmologies.

At $z = 1.80$ the source and CPTG comparison distances are

$$D_A^{\Lambda 70} = 1741.846\,\mathrm{Mpc}, \tag{52}$$

$$D_{A,\mathrm{ac}} = 1791.516\,\mathrm{Mpc}, \tag{53}$$

$$D_{A,\mathrm{lum}} = 1656.481\,\mathrm{Mpc}, \tag{54}$$

$$D_{A,\mathrm{mid}} = \tfrac{1}{2}(D_{A,\mathrm{ac}} + D_{A,\mathrm{lum}}) = 1723.998\,\mathrm{Mpc}. \tag{55}$$

Re-expressing R_e by $s_D = D_A/D_A^{\Lambda 70}$ and the constant-M/L stellar normalization by s_D^2 gives the following full projection sensitivity:

Distance coordinate	R_e (kpc)	$\eta_\star$	Unit-lock separation
Acoustic endpoint	1.68677	1.0073 ± 0.3614	0.020σ
Symmetric midpoint	1.62320	1.0468 ± 0.3756	0.125σ
Luminosity endpoint	1.55963	1.0894 ± 0.3909	0.229σ

Table 3: Distance-coordinate sensitivity. The quoted $\eta_\star$ uncertainties propagate the observed $\sigma_\star$ and R_e errors at fixed comparison coordinate. The acoustic-to-luminosity span is a deterministic projection sensitivity, not an additional statistical uncertainty.

The distance test leaves the central conclusion unchanged. Across the full current CPTG acoustic-to-luminosity distance span, the inferred ratio runs from 1.0073 ± 0.3614 to 1.0894 ± 0.3909. Exact unit locking remains only 0.020σ to 0.229σ from the corresponding central values. The near-unity result is therefore not produced by choosing the original ΛCDM distance convention or by selecting the symmetric midpoint of the CPTG projections.

8 Radial Field Reconstruction and Structural Mode

The native-locking result is a scalar comparison between the galaxy's structural acceleration and the native cosmological scale. The available S'ersic structure allows the analysis to proceed further and examine the radial organization of the solved field. In CPTG this downstream information is summarized by Structural Mode N, which measures where the curvature-polarization/transport structure changes most strongly across the galaxy [5].

This provides a second, distinct diagnostic of the same system. A model could reproduce a one-radius acceleration while organizing the radial field implausibly. Conversely, a structurally coherent N does not rescue a failed native-locking comparison. The two diagnostics probe different aspects of the same solved object.

For a solved radial acceleration field $g(r)$, CPTG first constructs the structural invariant

$$C_g(r) = g(r)^2 + \kappa_{\rm struct}^2 \left(\frac{dg}{dr}\right)^2, \qquad \kappa_{\rm struct} = 1\,\mathrm{kpc}, \tag{56}$$

The first term tracks field amplitude and the second tracks radial change over the fixed structural length $\kappa_{\rm struct} = 1\,\mathrm{kpc}$. Structural Mode does not classify the galaxy directly from C_g. It weights the places where this invariant itself changes most rapidly, using

$$w(r) = \left|\frac{dC_g}{dr}\right|. \tag{57}$$

The weighted centroid λ therefore marks the characteristic radius at which the retained structural change is concentrated. Dividing the registered outer radius by that organization length gives

$$\lambda = \frac{\int_{R/5}^{R} r\, w(r)\, dr}{\int_{R/5}^{R} w(r)\, dr}, \qquad N = \frac{R}{\lambda}. \tag{58}$$

This makes the interpretation intuitive: if the dominant structural change sits farther outward, λ is larger and N is smaller; if it is concentrated farther inward, λ is smaller and N is larger. It also makes clear why N_π and N remain distinct:

$$N_\pi: \text{ cosmological phase}, \qquad N: \text{ radial organization of the solved galaxy field.} \tag{59}$$

8.1 Stellar-Source Reconstruction from the Observed Sérsic Profile

The source study provides the Sérsic index and projected axis ratio needed to extend the one-radius calculation into a radial structural reconstruction. The calculation assumes, explicitly, that stellar mass follows the fitted F160W Sérsic light profile with spatially constant stellar mass-to-light ratio. The source analysis does not provide a measured radial gas-mass profile for NMBS-C7447. The present radial CPTG source is therefore treated as stellar dominated and normalized to the reported stellar mass $M_\star$; any unmeasured non-stellar baryonic component lies outside this Structural Mode reconstruction. For the observed profile, write

$$S(s) = \exp\left[-b_n\left(\left(\frac{s}{R_e}\right)^{1/n} - 1\right)\right], \tag{60}$$

with $n = 5.3$. Because the ideal Sérsic model has no finite outer edge, this paper registers the Structural Mode outer readout radius as $R \equiv R_{95}$, the projected radius enclosing 95 percent of the model light:

$$\frac{\gamma\Big(2n, b_n (R_{95}/R_e)^{1/n}\Big)}{\Gamma(2n)} = 0.95, \qquad \frac{\gamma(2n, b_n)}{\Gamma(2n)} = \frac{1}{2}. \tag{61}$$

The choice $R = R_{95}$ is a registration rule needed because an ideal Sérsic model extends indefinitely. It is not a claim that the observations directly resolve stellar mass to 20–24 kpc.

Later we vary the outer-light registration from 90 to 99 percent to show how strongly the returned mode depends on that modeling choice. The area-equivalent spherical reconstruction uses the published circularized radius directly, giving

$$R_{95,\mathrm{circ}} = 20.00054\,\mathrm{kpc}. \tag{62}$$

The spherical model is only a baseline because the galaxy is measurably flattened. For the axisymmetric reconstruction, the published circularized radius and projected axis ratio imply the semimajor-axis half-light radius

$$R_{e,\mathrm{maj}} = \frac{R_{e,\mathrm{circ}}}{\sqrt{q_{\mathrm{proj}}}} = 1.94632\,\mathrm{kpc}. \tag{63}$$

For similar oblate stellar-density surfaces,

$$m^2 = R^2 + \frac{z^2}{q_{\mathrm{int}}^2}, \qquad q_{\mathrm{proj}}^2 = \cos^2 i + q_{\mathrm{int}}^2 \sin^2 i, \tag{64}$$

so the measured $q_{\mathrm{proj}} = 0.71$ admits $0 < q_{\mathrm{int}} \leq 0.71$ with the edge-on endpoint $q_{\mathrm{int}} = 0.71$.

Let

$$L_S = 2\pi \int_0^\infty S(s)s\,ds. \tag{65}$$

Under the declared constant stellar mass-to-light reduction, the oblate density is

$$\rho_q(m) = -\frac{M_\star}{\pi q_{\mathrm{int}} L_S} \int_m^\infty \frac{dS/ds}{\sqrt{s^2 - m^2}}\,ds, \tag{66}$$

and the exact equatorial source field of nested similar oblate homoeoids is

$$g_{\mathrm{bar,obl}}(R) = \frac{4\pi G q_{\mathrm{int}}}{R} \int_0^R \frac{\rho_q(m) m^2\,dm}{\sqrt{R^2 - (1 - q_{\mathrm{int}}^2) m^2}}. \tag{67}$$

The $q_{\mathrm{int}} = 1$ limit reduces to the spherical source $GM_\star(< R)/R^2$. The edge-on case $q_{\mathrm{int}} = q_{\mathrm{proj}} = 0.71$ requires no new fitted normalization; thinner intrinsic shapes represent the allowed inclination/thickness family consistent with the same projected axis ratio.

At every radius the same observed-epoch structural estimator from Eq. (25) is used in the registered CPTG response law,

$$g = g_{\mathrm{bar}} + \frac{\sqrt{a_\star g_{\mathrm{bar}}} + g_{\mathrm{geom}}}{\left[1 + (g/a_\star)^{123/50}\right]^{11/50}}, \tag{68}$$

with

$$g_{\mathrm{geom}}(r) = \frac{1}{2} a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \qquad R_c = \frac{48\pi c^2}{a_\star}. \tag{69}$$

No morphology label, CSMI class, or native-locking residual is inserted into this radial solve. The calculation therefore asks the field to reveal its structural organization rather than instructing it what class of galaxy it should resemble.

8.2 Structural Mode Result and Geometry Dependence

The returned Structural Mode is stable across the central geometry choices considered here. The simplest area-equivalent spherical baseline gives

$$\lambda_{\rm sph,circ} = 6.51415\,{\rm kpc}, \qquad N_{\rm sph,circ} = 3.07032. \tag{70}$$

To isolate source flattening from the conversion between circularized and semimajor radii, a spherical control using the same semimajor-axis Sérsic scale gives

$$R_{95,\rm maj} = 23.73628\,{\rm kpc}, \qquad N_{\rm sph,maj} = 3.04944. \tag{71}$$

The more informative central result carries the measured flattening into the baryonic source. At the edge-on endpoint, without fitting any new source normalization, the solved field gives

$$\lambda_{\rm obl} = 7.74518\,{\rm kpc}, \qquad N_{\rm obl} = 3.06465. \tag{72}$$

As the intrinsic thickness decreases through the allowed inclination family, the central solution rises monotonically. The explicitly calculated thin-geometry stress point

$$q_{\rm int} = 0.005 \tag{73}$$

gives

$$N = 3.11552. \tag{74}$$

Continuing the numerical sequence to $q_{\rm int} = 2 \times 10^{-4}$ gives

$$N = 3.11592, \tag{75}$$

showing that the sequence is already near its thin-geometry asymptote. Equation (75) is a numerical near-thin limit, not an analytic $q_{\rm int} = 0$ solution.

Structural reconstruction	N	CSMI readout
Area-equivalent spherical baseline	3.07032	Early Spiral
Same-semimajor spherical geometry control	3.04944	Intermediate Spiral
Measured-flattening oblate edge-on endpoint	3.06465	Early Spiral
Thin-geometry stress point, $q_{\rm int} = 0.005$	3.11552	Early Spiral
Near-thin numerical point, $q_{\rm int} = 2 \times 10^{-4}$	3.11592	Early Spiral

Table 4: Post-solution Structural Mode results. CSMI is a downstream descriptive lookup inherited from the current Structural Mode catalog; continuous N is the scientific coordinate.

The numerical label is less important than what produced it. At fixed semimajor profile, representing the observed flattening shifts the mode from 3.04944 to 3.06465, and thinner allowed intrinsic shapes move the structural organization somewhat farther inward. Nothing about $\eta_\star$ is changed to obtain that result. The inherited CSMI table happens to call much of this interval "Early Spiral," but that is only a downstream catalog label for the value of N; it is not a claim that NMBS-C7447 is morphologically an ordinary early spiral. Continuous N is the scientific quantity.

The registered outer-light choice is also tested directly. For the same central measured-flattening edge-on source,

$$N(R_{90}) = 3.08076, \qquad N(R_{95}) = 3.06465, \qquad N(R_{99}) = 3.08343. \tag{76}$$

The continuous mode therefore remains near $N \simeq 3.07$ across a broad 90–99 percent Sérsic-light registration range. The exact number moves slightly, as it should when the outer readout changes, but the structural conclusion does not depend on choosing 95 percent specifically.

9 Uncertainty Structure, Sensitivity, and Falsifiability

The cross-scale and Structural Mode results are affected by observational uncertainty in different ways. Native locking is presently measurement limited. Because $a_\sigma \propto \sigma_\star^2$, the $\pm 51\,\mathrm{km\,s^{-1}}$ dispersion uncertainty dominates the propagated result, leaving $\eta_\star = 1.036 \pm 0.372$. A substantially more precise stellar dispersion would therefore provide the most direct improvement to the native-locking test.

Structural Mode behaves differently. Its central value is much less sensitive to any one quoted source perturbation because N measures the radial organization of the solved field rather than its absolute amplitude. For the measured-flattening edge-on endpoint, the one-at-a-time ranges are

Perturbed input	Returned N interval	Comment
$\sigma_\star = 294 \pm 51\,\mathrm{km\,s^{-1}}$	3.05682–3.07765	Pressure-support estimator changes
$R_e = 1.64 \pm 0.15\,\mathrm{kpc}$	3.05830–3.07275	Radius affects both support and source scale
$n = 5.3 \pm 0.4$	3.05769–3.07046	Radial concentration sensitivity
$\log_{10} M_\star \pm 0.2$ dex	3.05339–3.08142	Source-normalization sensitivity
$q_{\mathrm{proj}} = 0.71 \pm 0.01$	3.06448–3.06482	Edge-on geometry sensitivity

Table 5: One-at-a-time Structural Mode sensitivities at the measured-flattening edge-on endpoint. These are deterministic sensitivity intervals, not confidence intervals.

All listed one-at-a-time edge-on perturbations remain above the current $N = 3.05$ CSMI boundary. A linearized combination of the quoted $\sigma_\star$, R_e, n, and q_{proj} measurement scales gives an illustrative N scale of about 0.0133 around the central edge-on result. If the source paper's 0.2 dex stellar-mass systematic is treated as an additional symmetric scale, the corresponding linearized value is about 0.0181. Neither quantity is a formal confidence interval because a joint covariance model is unavailable and the stellar-mass term is systematic.

A deterministic simultaneous signed-corner scan is therefore also reported. At the measured-flattening edge-on endpoint, all quoted source extrema give

$$3.03026 \le N_{\mathrm{edge,corner}} \le 3.13356. \tag{77}$$

Allowing the same source corners to span the full edge-on to near-thin geometry family gives the deliberately conservative compound envelope

$$3.03026 \lesssim N_{\rm source+geom} \lesssim 3.21116. \tag{78}$$

These are deliberately conservative deterministic envelopes, not statistical confidence intervals. They show two things at once: the continuous Structural Mode remains a well-defined reconstruction, but a hard categorical CSMI label should not be treated as uncertainty-proof when all source and geometry extremes are compounded.

The native-locking coordinate and Structural Mode are also not statistically independent even though neither is used to tune the other. Both inherit $\sigma_\star$ and R_e. Under an illustrative independent-input linearization, including the 0.2 dex mass scale only in the N variance, the shared-input covariance gives

$$\rho_{\eta_\star,N} \simeq -0.39. \tag{79}$$

This number is diagnostic rather than inferential. It tells us that the clean conceptual separation between $\eta_\star$ and N does not imply statistical independence: both ultimately inherit some of the same measurements. Any future population relation $\eta_\star = \mathcal{F}(N)$ must therefore propagate shared-input covariance rather than count the two derived coordinates as independent data.

The native-locking formulation also admits a direct falsification criterion. For a set of independently measured bound systems, define

$$\delta_\star \equiv \ln \eta_\star = \ln\!\left(\frac{a_\star}{a_\pi}\right). \tag{80}$$

The strongest universal unit-lock hypothesis predicts $\delta_\star = 0$ after measurement uncertainty is accounted for. NMBS-C7447 can test that unit benchmark at its observed phase, but one object observed at one phase cannot by itself establish that $\eta_\star$ remains invariant as native phase changes. A broader structural-lock hypothesis permits an object-dependent but reproducible relation such as

$$\eta_\star = \mathcal{F}(N) \tag{81}$$

or a geometry-dependent extension. Those relations must be diagnosed from solved systems; they must not be fitted into the definition of $a_\star$ or N for the present object.

This paper does not adjudicate the distinct alternative CPTG formulation in which redshift selects only cosmological phase and the bound-system structural scale has no native-locking constraint. That formulation is mathematically separable and should be tested against the same observations in a dedicated companion analysis. Keeping the tests separate prevents either formulation from being favored by construction.

10 Physical Interpretation

10.1 Cross-Scale Consistency at the Observed Phase

NMBS-C7447 can be summarized by two CPTG coordinates that describe different aspects of its solved state. At its observed phase,

$$(\eta_\star, N_{\rm obl}) = (1.036 \pm 0.372,\ 3.06465), \tag{82}$$

for the measured-flattening edge-on source. The first number describes the placement of the galaxy's measured structural acceleration relative to the native cosmological acceleration. The second describes the radial organization of the solved field.

The principal cross-scale result is straightforward: the observed pressure-support acceleration is only 3.60% above the CPTG native acceleration at the same redshift. With the present observational error, that is a 0.10σ separation. The uncertainty is too large to establish exact equality, but the agreement is much closer than the error bar and survives the full current CPTG distance-coordinate sensitivity. The object therefore provides a well-defined discovery case in which the native-locking relation can be formulated and tested quantitatively.

Structural Mode adds a different piece of information. The measured-flattening solution gives $N = 3.06465$. One-at-a-time source perturbations span 3.05339–3.08142; the simultaneous edge-on source corner spans 3.03026–3.13356. The central mode therefore describes a stable radial organization even though the inherited categorical CSMI label can move under compounded extremes. The result is not being obtained by feeding a Hubble type, CSMI class, or other categorical morphology label into the solve. The measured Sérsic profile and axis ratio enter as source-structure observables, while N itself is read from the returned CPTG field.

10.2 Relation to Standard ΛCDM Inference

CPTG and ΛCDM begin from the same measured redshift, stellar velocity dispersion, and HST structural quantities. Their distinction in this paper lies in how those observations are organized within the gravitational and cosmological inference.

In the standard ΛCDM framework, redshift is placed on a Friedmann cosmological background and galaxy formation is ordinarily interpreted within a dark-matter-halo hierarchy. The compactness of NMBS-C7447 is then part of the broader problem of early assembly, quenching, merger history, and subsequent size growth [2, 1]. For NMBS-C7447 specifically, the source paper does not fit a dark-halo mass profile; its quoted dynamical mass is a Sérsic-dependent virial estimate built from the measured radius and stellar dispersion. This paper does not attempt to falsify the standard framework with one galaxy, and it does not compute a ΛCDM likelihood comparison.

CPTG instead separates the problem into native geometry and bound-system structure. The redshift selects one native π-branch phase. In the present pressure-supported reduction, the galaxy's own kinematics supply a one-radius estimate of its object-dependent structural acceleration $a_\star$. No universal MOND-like acceleration is imposed, and no dark-halo parameter is introduced to define that support scale. CPTG then asks whether the local structural scale and native cosmological scale possess a common geometric ratio, and separately whether the baryon-sourced solved field organizes itself coherently through Structural Mode N.

The CPTG inference sequence can therefore be summarized as

$$\begin{gathered}\text{observed redshift} \longrightarrow \text{native geometric phase},\\ \text{observed stellar support} \longrightarrow a_\star,\\ \text{baryonic source} \longrightarrow g_{\mathrm{CPTG}}(r) \longrightarrow N,\\ \text{only then:} \quad \eta_\star = \frac{a_\star}{a_\pi}.\end{gathered} \tag{83}$$

The significance of this construction depends on whether the same ordering continues to work across systems not used to motivate the present formulation. A repeatable $\eta_\star$ law would be a new cross-scale relation between bound structure and native cosmological geometry. A reproducible relation between $\eta_\star$ and N would be even more specific, because it would connect cosmological placement with the internally solved organization of the galaxy without using one to fit the other.

10.3 Scope and Evidentiary Status

For NMBS-C7447, the present calculation is consistent with the unit-lock benchmark: exact unit locking lies comfortably inside the observational uncertainty, and the radial field reconstruction returns a stable continuous Structural Mode. But NMBS-C7447 was also the object on which the native-locking formulation was recognized and clarified. It is therefore a discovery and derivation case, not an independent prospective validation of a universal law.

The evidentiary question is therefore transferred to the next stage of testing. The same reduction must be applied, without retuning the native branch or redefining $a_\star$, to a broader set of pressure-supported galaxies spanning independent native phases. If $\eta_\star$ scatters without reproducible structure, native locking fails. If it remains stable across systems with different masses, sizes, Structural Modes, and redshift-selected native phases, the result becomes much harder to dismiss as coincidence. If instead $\eta_\star$ varies systematically with N or source geometry, then the simple unit-lock picture would be replaced by a more informative structural relation rather than assumed in advance. The decisive phase test is residual: after any reproducible dependence on N and source geometry is accounted for, a genuine native-locking relation should not require an additional systematic dependence on N_π merely to remain consistent across the sample.

11 Conclusion

NMBS-C7447 is an unusually informative galaxy because the observations connect an early cosmological epoch to a directly measured compact stellar structure. The source data show a $\sim 1.5 \times 10^{11} M_\odot$ quiescent galaxy with $R_e = 1.64 \pm 0.15\,\mathrm{kpc}$ and $\sigma_\star = 294 \pm 51\,\mathrm{km\,s^{-1}}$, structurally much denser than comparable low-redshift systems. Those same measurements allow CPTG to ask a cross-scale question that is not part of the original observational analysis.

The central inference follows directly from the separation of cosmological and bound-system information. Redshift fixes the native CPTG phase and gives

$$a_\pi(z = 1.80) = 1.64863 \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{84}$$

The galaxy independently gives

$$a_\star \simeq a_\sigma = (1.708 \pm 0.613) \times 10^{-9}\,\mathrm{m\,s^{-2}}. \tag{85}$$

Their central values differ by only 3.60%, corresponding to 0.10σ relative to the propagated observational uncertainty. In dimensionless form,

$$\eta_\star = \frac{R_C}{R_c} = \frac{a_\star}{a_\pi} = 1.036 \pm 0.372. \tag{86}$$

The near-unit result persists when the published distance-dependent quantities are carried across the current CPTG acoustic-to-luminosity comparison span, so it is not created by one distance convention.

The radial reconstruction then asks a separate question. Using the measured Sérsic structure and flattening, the post-solution CPTG field gives

$$N_{\mathrm{obl}} = 3.06465, \tag{87}$$

with one-at-a-time source sensitivities 3.05339–3.08142, a simultaneous edge-on source corner of 3.03026–3.13356, and a near-thin geometry sequence approaching $N \simeq 3.1159$ for the central source. Structural Mode therefore adds information about how the solved field is organized rather than merely repeating the one-radius acceleration result.

The broader significance lies in the inference architecture. ΛCDM and CPTG can begin from the same measured galaxy while organizing the inference differently. The standard framework places the galaxy inside an expanding Friedmann background and a dark-matter halo-based structure-formation picture. CPTG places the observation on one native geometric phase, derives an object-dependent structural acceleration from the bound system itself, solves the baryon-sourced polarization/transport field, and only afterward compares local and cosmological geometry. For NMBS-C7447, that independently assembled comparison lands unexpectedly close to the strongest unit-lock benchmark.

A single object cannot establish a universal law, but it can define a precise prospective test. The native-locking hypothesis should survive only if the same predeclared construction continues to work across independent pressure-supported systems spanning native phase, with observational uncertainties and shared-input covariance carried explicitly. That population test will determine whether the 3.60% agreement found here is coincidence, a mode-dependent structural relation, or evidence of a genuine cross-scale CPTG invariant.

Data, Code, and Evidence Availability

The observational authority for NMBS-C7447 is van de Sande et al. [1]. The standard ΛCDM comparison language is used only for contextual contrast and for documenting the source paper's published distance convention; the governing CPTG cosmological construction is the native π branch. The native CPTG branch, structural acceleration hierarchy, and local/native curvature-radius definitions follow the current unified framework and π-branch comparison-coordinate paper [3, 4]. Structural Mode and the downstream CSMI lookup follow the current primary-SPARC Structural Mode validation [5].

The exact numerical evidence authority for the native-phase calculation, pressure-support propagation, native-locking ratio, distance-coordinate sensitivity, Sérsic and oblate source reconstruction, nonlinear field solve, Structural Mode evaluation, source and geometry sensitivity scans, and shared-input covariance diagnostic is archived as one self-contained evidence package. The archive contains the authoritative numerical script and output together with a SHA-256 manifest, reproduction instructions, execution-environment record, and a clean rerun verification record. It also verifies monotonicity of $C_g(r)$ over the registered Structural Mode interval for the central and sensitivity solves and cross-checks the endpoint evaluation against direct integration of $|dC_g/dr|$.

`CPTG_NMBS-C7447_NativeLock_Structural_Evidence_20260904_r04_FINAL_EVIDENCE.zip`

`SHA-256: fc3076d3ee5a83fd17f1a1a75fe73844bb36974440a0a204d7009c446247420a`

The public CPTG source repository is available through the companion resource [6].

11.1 Companion Mathematical Comparison

A comparison experiment was also carried out with this NMBS-C7447 analysis:

NMBS-C7447 at z=1.80 in CPTG: Native-Phase Placement, Local Structural Invariance, and Structural Mode N.

Because both papers begin from the same observed NMBS-C7447 measurements and return the same observed-state CPTG quantities, the comparison used that common solution as the starting point and examined only the different mathematical treatment of the galaxy away from its measured native phase. The experiment compared the behavior of $a_\star$, R_c, the structural-to-native acceleration ratio $a_\star/a_\pi$, the reconstructed CPTG radial field, and Structural Mode N under the two prescriptions.

The purpose was to determine whether native locking and local structural invariance are mathematically equivalent descriptions or genuinely different native-phase trajectories, and to identify precisely where their predictions separate without altering the definitions used in either paper. The full derivation, numerical comparison, and interpretation are presented in: *NMBS-C7447 at $z = 1.80$ in CPTG: Native Locking and Local Structural Invariance Compared.*

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